

Introduction

Pengyang Shao received his Ph.D. degree in July 2025 from Hefei University of Technology under the supervision of Prof. Meng Wang. As a Ph.D. candidate, his research interests were related to information retrieval. He is currently a postdoc at NExT++ Lab, NUS, under the supervision of Prof. Tat Seng Chua, focusing on LLM unlearning/safety.

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Research Summary

Pengyang Shao is also enthusiastic about expanding into new AI application domains beyond his current research areas, rather than limiting himself to a fixed set of applications. Regardless of the specific domains, his central research interest remains the same: understanding whether AI systems can be developed, evaluated, and deployed in a trustworthy, reliable, safe, and responsible manner, e.g., studying how LLM can remove sensitive knowledge and resist harmful behaviors.

Selected Publications

*** denotes the corresponding author, † denotes equal contribution.**

Under Review:

- Enyi Shi, **Pengyang Shao**, Yanxin Zhang, Chenhang Cui, Jiayi Lyu, Xiaobo Xia, Fei Shen, Tat-Seng Chua. Lingua-safetybench: A Benchmark for Safety Evaluation of Multilingual Vision-Language Models. Submitted to IEEE Transactions on Information Forensics and Security (TIFS).
- **Pengyang Shao**, Yanzheng Jin, Fei Shen, Kenji Kawaguchi, Xun Yang, Meng Wang. UniBal: A Unified Balancing Framework for Gradient-based LLM Unlearning. Submitted to IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI).
- **Pengyang Shao**, Chuanpeng Lu, Wei Qin, Yanzheng Jin, Xiaohao Liu, Xi Ai, Kenji Kawaguchi, Richang Hong. Beyond Cross-Lingual Transfer: Benchmarking Propagation Boundaries in Multilingual LLM Unlearning. Submitted to KDD 2027 August Datasets and Benchmarks Track.
- Yanzheng Jin, **Pengyang Shao**, Xiaohao Liu, Xi Ai, Fei Shen, Kenji Kawaguchi. CoLAS: Multimodal Corroboration of Latent Asset Signals for Financial Trading. Submitted to KDD 2027 August Research Track.
- Chao Chen, **Pengyang Shao**, Wei Qin, Shijie Hao, Fei Shen, Richang Hong. Revisiting Over-refusal in Safety-aligned Large Language Models: A Representation and Optimization Perspective. Submitted to KDD 2027 August Research Track.
- Enyi Shi, Fei Shen, Shuyi Miao, Linxia Zhu, **Pengyang Shao**, Jinhui Tang, Tat-Seng Chua. Precise Shield: Explaining and Aligning VLLM Safety via Neuron-Level Guidance. Submitted to KDD 2027 August Research Track.

Published:

- Fengbin Zhu[†], Zijing Cai[†], Yuzhe Wang, **Pengyang Shao***, Wenjie Wang, Fuli Feng, Richang Hong*, Tat-Seng Chua. MURE: Hierarchical Multi-Resolution Encoding via Vision-Language Models for Visual Document Retrieval. ACM International Conference on Multimedia Retrieval (ICMR 2026).
- Naixin Zhai, **Pengyang Shao***, Binbin Zheng, Yonghui Yang, Fei Shen, Long Bai, Xun Yang*. Maximizing Local Entropy Where It Matters: Prefix-Aware Localized LLM Unlearning. The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026) Outstanding Paper Award. [\[PDF\]](#) [\[Code\]](#)
- **Pengyang Shao**, Naixin Zhai, Lei Chen, Yonghui Yang, Fengbin Zhu*, Xun Yang*, Meng Wang. BalDRO: A Distributionally Robust Optimization based Framework for Large Language Model Unlearning. The ACM Web Conference (WWW 2026) Oral Presentation. [\[PDF\]](#) [\[Code\]](#)
- **Pengyang Shao**, Le Wu, Kun Zhang*, Lei Chen, Meng Wang*. Privacy Matters: Data Attack for Unlearnable User Preference in Recommendation. ACM Transactions on Information Systems (TOIS) 2026. [\[PDF\]](#) [\[Code\]](#)
- **Pengyang Shao**, Lei Chen*, Fei Liu, Yonghui Yang, Xun Yang, Meng Wang*. Multi-Agent Debate based Concept Augmentation for Enhanced Cognitive Diagnosis. SIGKDD Conference on Knowledge Discovery and Data Mining (KDD 2026) Poster Presentation. [\[PDF\]](#) [\[Code\]](#)
- Jilong Liu, **Pengyang Shao***, Wei Qin, Fei Liu, Yonghui Yang, Richang Hong*. Debate over Mixed-knowledge: A Robust Multi-Agent Framework for Incomplete Knowledge Graph Question Answering. The 40th Annual AAAI Conference on Artificial Intelligence (AAAI 2026) Oral Presentation. [\[PDF\]](#) [\[Code\]](#)
- **Pengyang Shao[†]**, Yonghui Yang[†], Chen Gao, Lei Chen, Kun Zhang, Chenyi Zhuang, Le Wu, Yong Li, Meng Wang*. Exploring Heterogeneity and Uncertainty for Graph-based Cognitive Diagnosis Models. SIGKDD Conference on Knowledge Discovery and Data Mining KDD 2025 Oral Presentation. [\[PDF\]](#) [\[Code\]](#)
- **Pengyang Shao**, Kun Zhang*, Chen Gao*, Lei Chen, Miaomiao Cai, Le Wu, Yong Li, Meng Wang. Breaking Student-concept Sparsity Barrier for Cognitive Diagnosis. Frontiers of Computer Science, 2025.
- **Pengyang Shao**, Le Wu*, Kun Zhang, Defu Lian, Richang Hong, Yong Li, Meng Wang*. Average User-side Counterfactual Fairness for Collaborative Filtering. ACM Transactions on Information Systems (TOIS) 2024. [\[PDF\]](#)
- **Pengyang Shao**, Le Wu*, Lei Chen, Kun Zhang, Meng Wang*. FairCF: Fairness-aware Collaborative Filtering. Science China Information Sciences, 2022. [\[PDF\]](#) [\[Code\]](#)

Experiences

1) National University of Singapore Advisor:	Postdoc Tat Seng Chua	2025-Present Singapore
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2) Hefei University of Technology	Ph.D.	2019 – 2025
Advisor:	Meng Wang	Hefei, China
3) Tsinghua University - FIB	Intern	2023 - 2024
Co-Advisor:	Yong Li	Beijing, China
4) Alibaba-Ant Group	Intern	2024
Advisor:	Chenyi Zhuang	Hangzhou, China

Academic Services

Organizers of KDD 2026 workshop: Personal Intelligence in the Agentic AI Era

Journal/Conference Reviewer: IEEE TKDE/TBD, ACM ToRS/TOIS/TIST/TiiS, npj Health Systems, Pattern Recognition, KDD, WWW, IJCAI, ACL, AAAI, CIKM, SIGIR, MM.

Academic Mentorship

1. Mentoring Yanzheng Jin, a PhD candidate at the National University of Singapore (NUS), on two research projects targeting IEEE TPAMI and KDD 2027, respectively, focusing on LLM unlearning and AI for finance.
2. Mentoring Jilong Liu, a PhD candidate at Hefei University of Technology (HFUT), on research in agent-based knowledge graph question answering for a paper on AAAI 2026, with contributions to methodology development, and manuscript preparation.
3. Mentoring Naixin Zhai, a master's student at the University of Science and Technology of China (USTC), on an LLM unlearning project for a paper on ACL 2026, providing guidance throughout the research process, from model development and experimental design to manuscript revision.
4. Mentoring Chao Chen, a PhD candidate at Hefei University of Technology (HFUT), on two research projects targeting ACM TOIS and KDD 2027, respectively, focusing on LLM personalization and LLM safety.

Awards

- 2026 [Outstanding Paper Award at ACL 2026](#)
- 2025 Best Paper Award at TIME 2025 (Workshop on ACM WWW 2025)
- 2024 [China Association for Science and Technology Young Talent Support Program](#)
- 2021-2024 First-Class Graduate Academic Scholarship (awarded annually)
- 2019 Outstanding Graduate of Anhui Province
- 2018 Mathematical Contest in Modeling - Meritorious Winner
- 2017 China Undergraduate Mathematical Contest in Modeling (CUMCM) National Second Prize